Enhanced Electron Nuclear control leveraging hyperfine coupling and radio-frequency drive

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Abstract

Solid-state defect systems have emerged as promising platforms for quantum information processing and sensing. In particular, the nitrogen-vacancy (NV) center in diamond, coupled with surrounding nuclear spins, exhibits long coherence times and enables high-fidelity operations. To implement high-fidelity, selective-qubit operations, dynamic decoupling (DD) sequences, consisting of multi-pulse control on the NV spin, have been developed to achieve entangling gates with selected target nuclear spins while reducing decoherence from the broader spin bath. On the other hand, an additional selective phase-controlled radio-frequency (rf) driving on nuclear spins, referred as DDrf gate, enables elaborated control of individual spins. The mechanisms underlying these approaches differ that dynamic decoupling relies on transverse hyperfine interaction components, whereas the DDrf gate exploits rf driving and suppresses transverse hyperfine coupling. In this work, we propose a hybrid DDrf gate that leverages both mechanisms to achieve faster and more robust gate operations. The hybrid gate achieves faster and more robust operations than either the DD or DDrf approach alone, and extends high-fidelity control to nuclear spins with weak transverse hyperfine coupling.

Nitrogen-Vacancy (NV) center system

In the nitrogen-vacancy (NV) center, such an operation can be implemented by a sort of Hamiltonian engineering. We consider a nitrogen-vacancy (NV) center electron spin ($S = 1$) coupled to surrounding ^{13}C nuclear spins ($I = \frac{1}{2}$) via the hyperfine interaction [2]. The Hamiltonian is

$$H = |0\rangle\langle 0| \otimes \underbrace{\omega_0 I_z}_{=H_0} + |1\rangle\langle 1| \otimes \underbrace{(\omega_0 - A_\parallel)\, I_z + A_\perp\, I_x}_{=H_1}$$

where ω_0 is the nuclear Larmor frequency, and $A_\parallel$, $A_\perp$ are the parallel and perpendicular hyperfine components. We can evaluate the time evolution of the ^{13}C nuclear spin depending on the NV center electron spin state in a piecewise manner.

Conditional Gate

The operation applied to the target depends on the state of the control qubit, and can be written as:

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1,$$

where U_0 and U_1 are single-qubit rotations conditioned on the control qubit state.

Dynamic Decoupling (DD)

We can enable conditional interaction by applying a dynamic decoupling sequence consisting of N sequential π-pulses. Consider the basic decoupling unit on the electron spin τ-π-2τ-π-τ where τ is a free evolution time.

CPMG Pulse Sequence

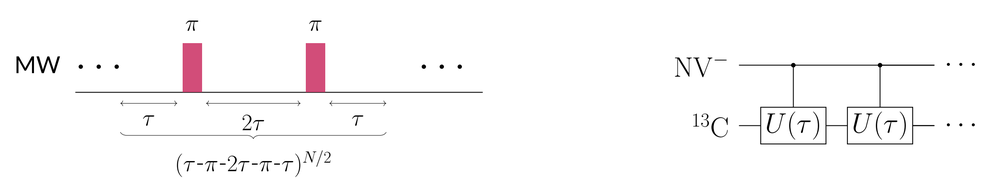

Figure 1. CPMG dynamical decoupling sequence on the NV electron spin (left) and equivalent quantum circuit (right).

The net result of this unit sequence results in a rotation by an angle ϕ around an axis $\vec{\sigma}_i$, depending on the initial state of the electron spin; $\vec{\sigma}_0$ for $m_s = 0$ state, and $\vec{\sigma}_1$ for $m_s = -1$ state. If $\vec{\sigma}_0$ and $\vec{\sigma}_1$ are not parallel, the resulting conditional rotation of the nuclear spin generally entangles the electron and nuclear spins. A dynamical decoupling sequence with the basic unit τ-π-2τ-π-τ, repeated $N/2$ times, makes further conditional operation.

Conditional Rotation & Resonance

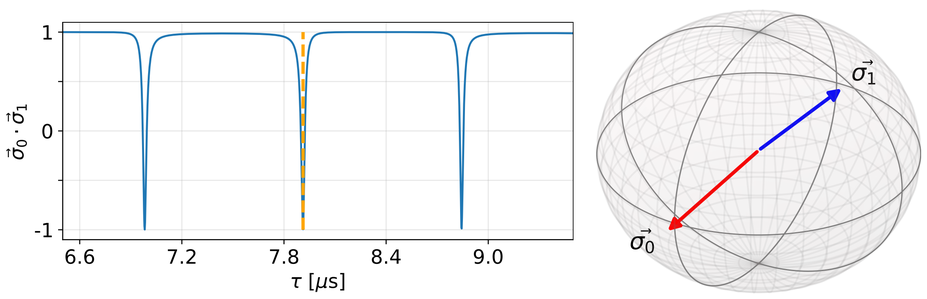

Figure 2. (Left) Inner product $\vec{\sigma}_0 \cdot \vec{\sigma}_1$ as a function of τ, showing sharp resonances where the axes become antiparallel; the orange dashed line marks the chosen τ_k. (Right) Conditional rotation axes $\vec{\sigma}_0$ (red) and $\vec{\sigma}_1$ (blue) on the Bloch sphere at resonance.

The conditional evolution operators for a single decoupling unit are:

$$U_0 = e^{-i\tau H_0} e^{-i2\tau H_1} e^{-i\tau H_0} = e^{-i\phi_{\mathrm{DD}}\vec{\sigma}_0} \tag{1}$$
$$U_1 = e^{-i\tau H_1} e^{-i2\tau H_0} e^{-i\tau H_1} = e^{-i\phi_{\mathrm{DD}}\vec{\sigma}_1} \tag{2}$$

Here $\omega_0 \equiv \omega_0$ and $\omega_1 = \sqrt{(\omega_0 + A_\parallel)^2 + A_\perp^2}$ are the nuclear precession frequencies for $m_s = 0$ and $m_s = -1$, respectively. The conditional rotation angle ϕ_{DD} and inner product between the two rotation axes are determined by the choice of τ and NV center system configuration.

$$\cos\phi_{\mathrm{DD}} = \cos\omega_1\tau\cos\omega_0\tau - \frac{\omega_0 - A_\parallel}{\omega_1}\sin\omega_1\tau\sin\omega_0\tau \tag{3}$$
$$1 - \vec{\sigma}_0 \cdot \vec{\sigma}_1 = \frac{A_\perp^2}{\omega_0^2}\frac{(1-\cos\omega_1\tau)(1-\cos\omega_0\tau)}{1+\cos\phi_{\mathrm{DD}}} \tag{4}$$

DDrf Gate

The DDrf gate combines dynamical decoupling on the electron spin with phase-controlled radio-frequency (rf) driving of the nuclear spin [1].

Hamiltonian

In the interaction picture and rotating frame with respect to the electron energy splitting and neglecting non-secular terms, the Hamiltonian of the NV-^{13}C system under the approximation $A_\perp \approx 0$ and $\Omega_{\mathrm{RF}} \ll (\omega_0 - \omega_1)$ is given

$$H = |0\rangle\langle 0| \otimes \underbrace{(\omega_0 - \omega_1) I_z}_{=H_0} + |1\rangle\langle 1| \otimes \underbrace{\Omega_{\mathrm{RF}}(\cos\phi I_x + \sin\phi I_y)}_{=H_1}. \tag{5}$$

where the final term describes the interaction of the nuclear spin with a radio frequency (RF) driving field polarized along the x axis (in the same direction as the hyperfine coupling), phase ϕ and Rabi frequency Ω_{RF}.

Pulse Sequence

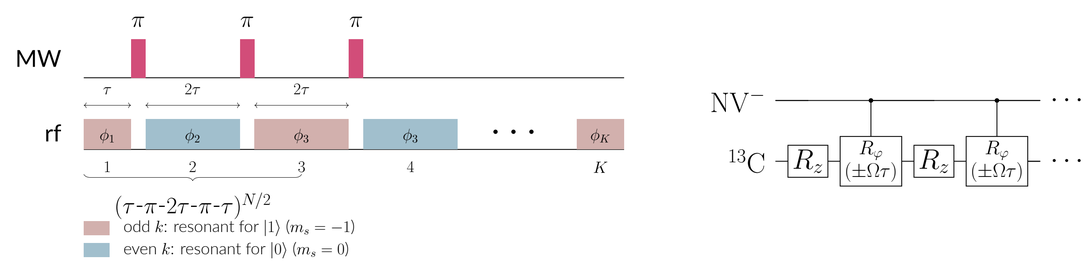

Figure 3. DDrf pulse sequence: MW π-pulses interleaved with phase-controlled rf pulses (left); equivalent quantum circuit showing the conditional $R_\varphi(\pm\Omega\tau)$ rotation per unit (right).

The electron spin state determines which rf pulses are resonant: odd-k pulses drive the nuclear spin when the electron is in $|1\rangle$ ($m_s = -1$), while even-k pulses are resonant for $|0\rangle$ ($m_s = 0$). The rf pulse phases $\phi_k = \varphi + \phi'_k$ are set to accumulate constructive rotations:

$$\phi'_k = \begin{cases} (k-1)\phi_\tau + \pi, & k \text{ odd} \\ (k-1)\phi_\tau, & k \text{ even} \end{cases}, \qquad \phi_\tau = A_\parallel \tau.$$

Controlled Rotation

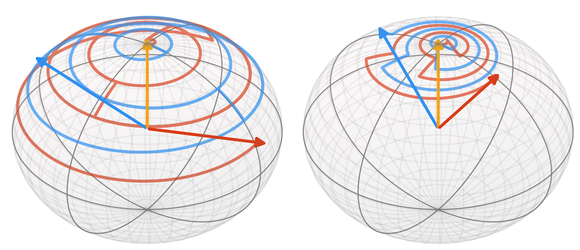

Figure 4. Nuclear spin Bloch sphere trajectories conditioned on the electron state (blue: $m_s = 0$, red: $m_s = -1$). (Left) Low $A_\perp$: the DDrf gate produces high-fidelity conditional rotation. (Right) High $A_\perp$: transverse hyperfine coupling distorts the trajectories, reducing gate fidelity.

The resulting two-qubit gate is a conditional rotation for each unit: where

$$U_0 = R_z(N(\omega_0 - \omega_1)\tau) R_\varphi(N\Omega_{\mathrm{rf}}\tau) \tag{6}$$
$$U_1 = R_z(N(\omega_0 - \omega_1)\tau) R_\varphi(-N\Omega_{\mathrm{rf}}\tau) \tag{7}$$

where $R_\varphi(\theta) = e^{-i\theta[\cos\varphi I_x + \sin\varphi I_y]}$.

Hybrid DDrf Gate

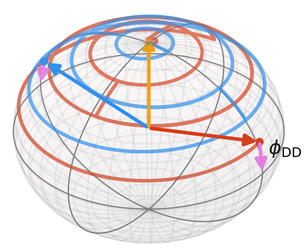

Figure 5. Bloch sphere trajectories of the hybrid DDrf gate. The combined DD and rf contributions produce a larger conditional rotation angle ϕ_D (pink arrow), enabling faster operations than either mechanism alone.

The transverse hyperfine interaction ($A_\perp$) is exploited by the DD gate, and the rf driving (Ω) by the DDrf gate. By allowing both contributions to act simultaneously, the hybrid approach can achieve:

- **Faster gate operations** through constructive combination of both driving mechanisms.
- **Broader applicability**, achieving high fidelity regardless of spin configuration.

To preserve the conditional operation arising from $A_\perp$ and the choice of τ, we choose τ at the DD resonance condition $\vec{\sigma}_0 \cdot \vec{\sigma}_1 \simeq -1$, i.e., $\tau = \frac{(2k+1)\pi}{\omega_0 + \omega_1}$, $\quad k = 0, 1, 2, \ldots$. In the regime of small $A_\perp$, the two conditional rotation angles combine additively, resulting in conditional gate:

$$U_0 = R_x(N(\phi_{\mathrm{DD}} + \Omega_{\mathrm{rf}}\tau)) \tag{8}$$
$$U_1 = R_x(-N(\phi_{\mathrm{DD}} + \Omega_{\mathrm{rf}}\tau)) \tag{9}$$

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[2] T. H. Taminiau, J. J. T. Wagenaar, T. van der Sar, F. Jelezko, V. V. Dobrovitski, and R. Hanson. Detection and control of individual nuclear spins using a weakly coupled electron spin. *Physical Review Letters*, 109:137602, 2012.